

EDWARD PRASCHIFKA

Graduate Software Engineer

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Graduate software engineer with a Bachelor of Mathematical and Computer Sciences, bringing ground-up technical depth together with real-time systems experience from an embedded software engineering internship at a defence technology company. Seeking an entry-level software engineering role.

SKILLS

Programming Languages	Python, C, C++, JavaScript
Frameworks & Databases	React, Express, Node.js, MongoDB, Flask, Django
ML & AI	Machine learning fundamentals, NumPy, Pandas
Math and Statistics	Linear algebra, multivariable calculus, probability, statistics
Tools & Practices	Git & collaborative workflows, test-driven development, Linux, AWS

PROJECTS

Orbit *Apr 2026 – May 2026*

- Built a full-stack pipeline that pulls raw data from the Steam Web API, transforms it, and serves it as an interactive 3D force-directed graph of a user's friend network.
- Designed a modular Flask backend (routes, services, graph, cache, metrics) with in-memory caching and rate limiting to manage API load and keep data flow efficient.
- Built incremental graph expansion, turning raw external API data into a clear, structured, navigable dataset.

MNIST Classifier — MLP from Scratch *May 2026 – Jul 2026*

- Built a feedforward neural network from first principles in NumPy — forward propagation, backpropagation, and gradient descent — without high-level ML frameworks, developing a ground-up understanding of how models actually learn.
- Built configurable CLI training and evaluation tools with tunable hyperparameters (batch size, learning rate, epochs) to support systematic experimentation.
- Achieved >97% accuracy on the MNIST 0-9 digit recognition dataset.

Bonsoir *Dec 2023 - Feb 2024*

- Built a full-stack voice assistant integrating Deepgram's speech recognition, ElevenLabs voice synthesis, and OpenAI's ChatGPT API to enable real-time spoken conversation with an AI assistant.
- Developed a React dashboard supporting chat creation, renaming, deletion, and message editing, backed by an Express REST API and MongoDB for persistent conversation storage.
- Deployed and ran the application on an AWS EC2 instance, gaining hands-on experience with cloud infrastructure and production deployment.

EXPERIENCE

Embedded Software Engineer Intern — Silentium Defence, Adelaide *Aug 2024 – Dec 2024*

- Conducted a cost-benefit analysis comparing two candidate solutions for sending real-time location data from a drone, weighing each against task requirements to inform the team's decision.
- Authored a comprehensive setup report enabling easy replication of the solution by other team members.
- Configured a Raspberry Pi with required OS, software, and hardware to reliably forward location data to endpoints.

EDUCATION

ANU China Studies Fellowship — Xi'an Jiaotong-Liverpool University, Suzhou, China *Feb 2025 – Feb 2026*

Bachelor of Mathematical & Computer Sciences — The University of Adelaide *Jan 2020 – Dec 2024*

Diploma of Languages (Mandarin) — The University of Adelaide *Jan 2020 – Dec 2024*

LANGUAGES & AWARDS

- Mandarin — HSK 5 (~2,500+ vocabulary), government-standard proficiency exam; used effectively in professional and academic contexts.
- CIW China Studies Fellowship (Feb 2025 – Feb 2026) — fully immersive year-long study program in China for Australian graduates.

VOLUNTEERING

- Kitchen Assistant, Meals on Wheels (Aug 2022 – Aug 2024) — prepared and delivered meals to hundreds of elderly clients as part of a small team.